Subject- Object Oriented Programming (CO212)
Unit 1 – Fundamentals of OOP
Topic – 1.3 Object Oriented Programming Paradigm, Basic Concepts of Object-Oriented Programming

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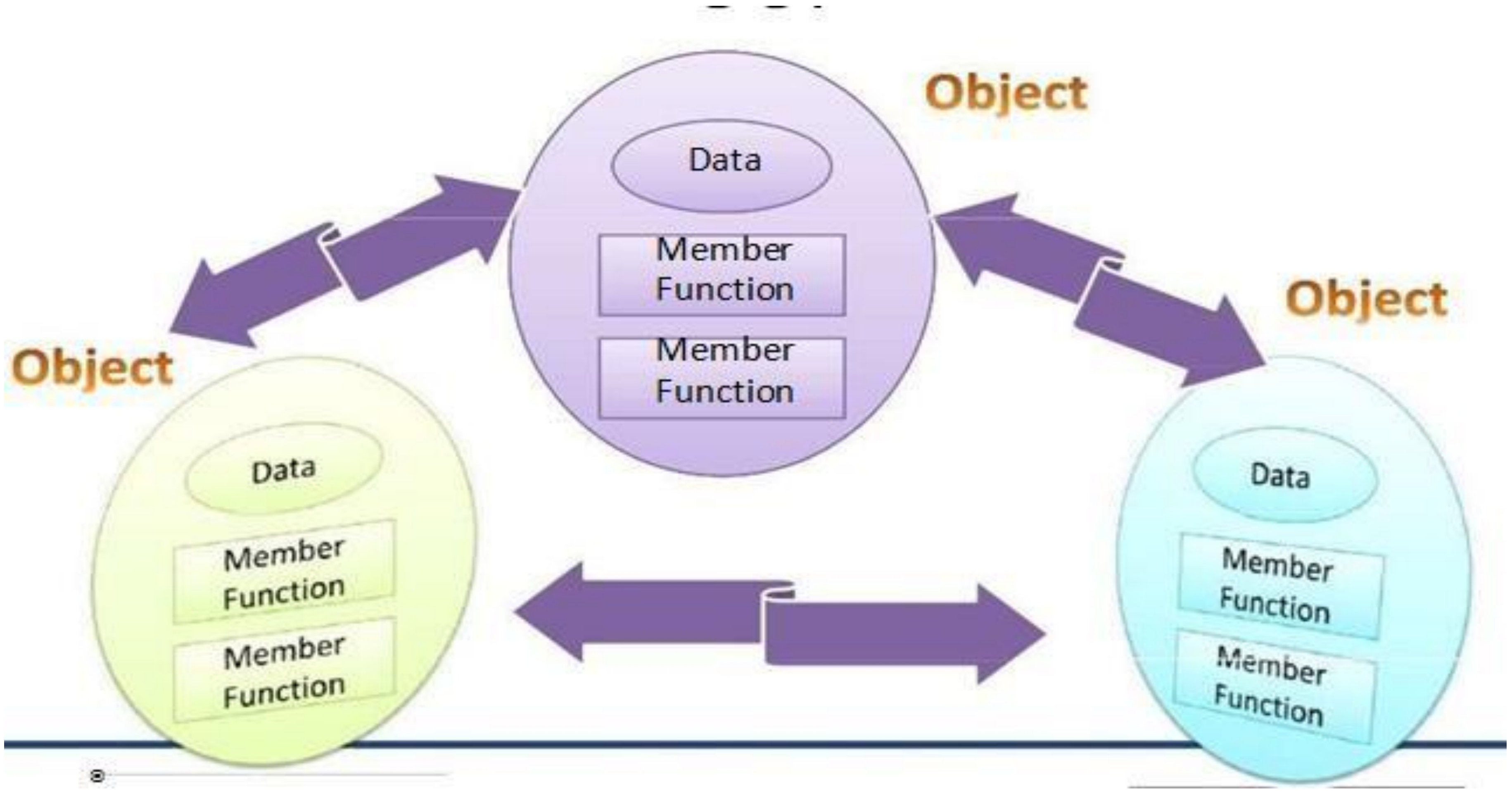
OOP paradigm

In OOP paradigm,

- Emphasis is on objects rather than procedure.
- Programs are divided into entities known as classes of objects.
- Data Structures are designed such that they characterize objects.
- Functions that operate on data are tied together in a class.
- Data is hidden and cannot be accessed by external functions.
- Objects communicate with each other through functions.
- New data and functions can be easily added whenever necessary.
- Follows bottom up approach in program design.

Organization of Data and Functions in OOP

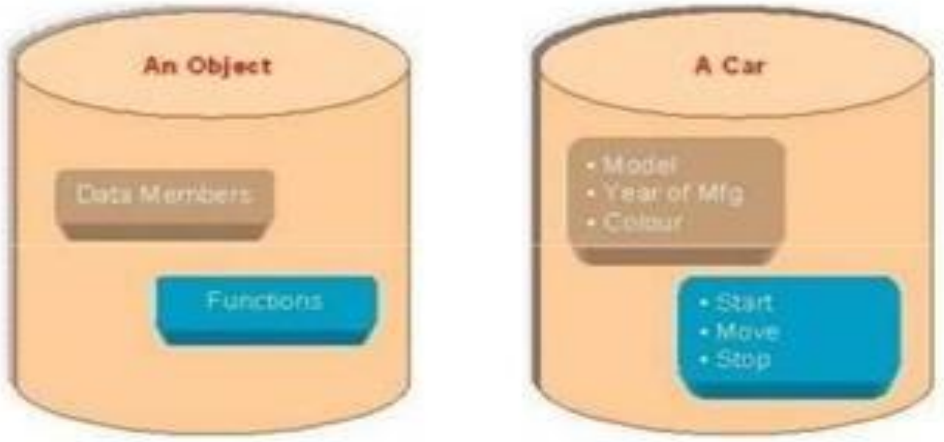


Basic Concepts in OOP

- Object
- Class
- Message Passing
- Abstraction
- Encapsulation
- Inheritance
- Polymorphism
- Dynamic Binding

Object

- Object is a basic run-time entity of a class.
- Object is also called as an instance of a class.
- It may represent a person, a place or any item that the program must handle.
- Example



Representation of an object

Class

- A class defines the type of an object.
- A class is a prototype/template or a blue print of an object.
- A class contains data and functions.
- Any number of objects can be created for a single class.
- We can create object of a class using syntax,

ClassName objectName;

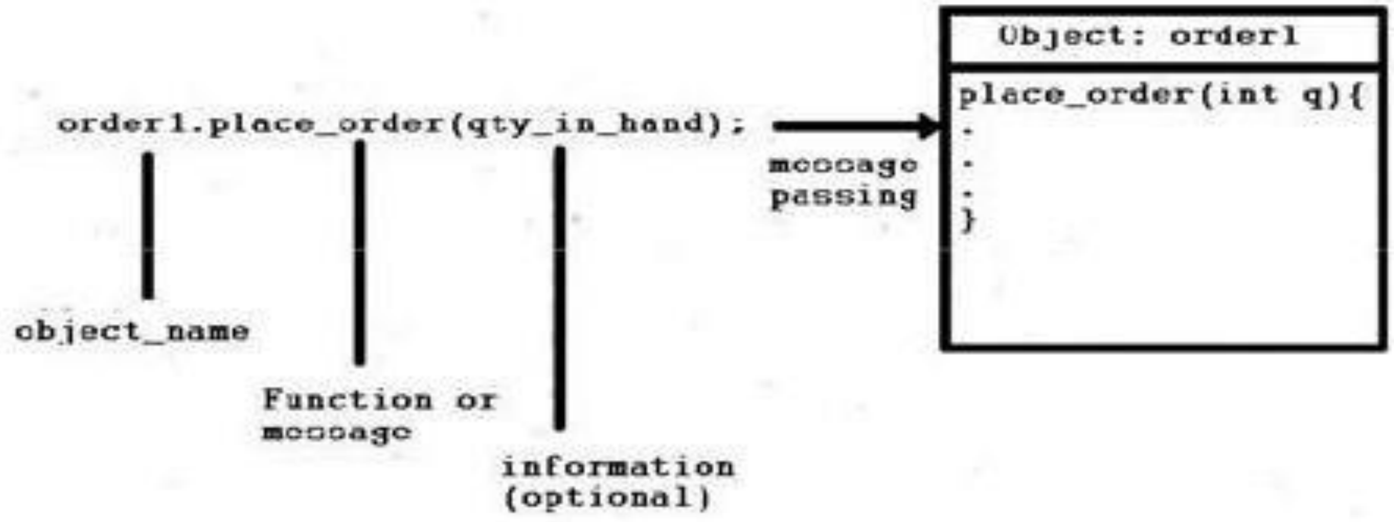
- Here ClassName is a class which is already defined and objectName is any user-defined name.
- For example, if Student is a class,

    Student neha,pratik;

In above example, neha and pratik are objects of class Student.

Message Passing

- A message passing is a request for execution of a procedure or to invoke a function in the receiving object that generates the desired result.
- Message passing involves specifying the name of the object, the name of the function i.e. message or the information to be sent.
- Objects can communicate with each other by passing messages same as people pass messages to each other.



Abstraction

- Abstraction refers to the representation of necessary features without including details.
- Classes use the concept of abstraction and are defined as a list of abstract data and functions to operate on these attributes.
- Data abstraction is a programming technique that relies on the separation of interface and implementation.

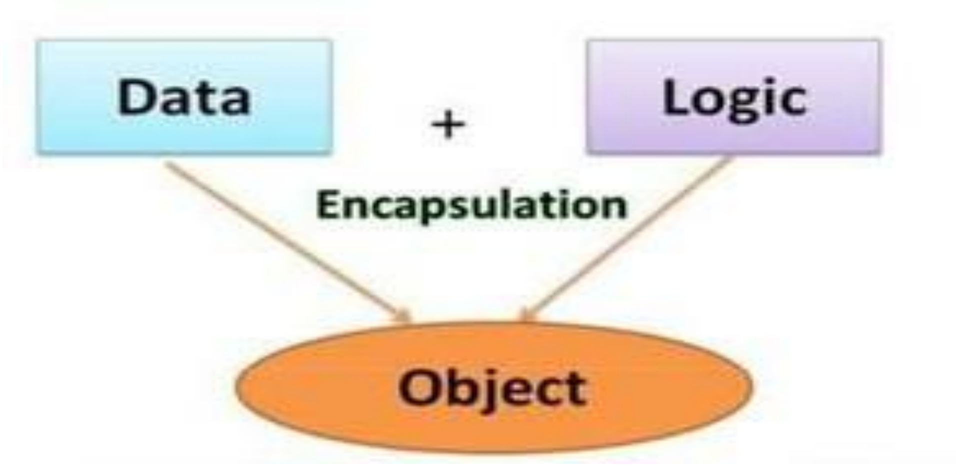
Abstraction

- When you press a key on your keyboard the character appears on the screen, you need to know only this, but How exactly it works electronically, is not needed.

- Another Example is when you use the remote control of your TV, you do not bother about how pressing a key in
the remote changes the channel on the TV.
  You just know that pressing
  the + volume button will increase the
volume.

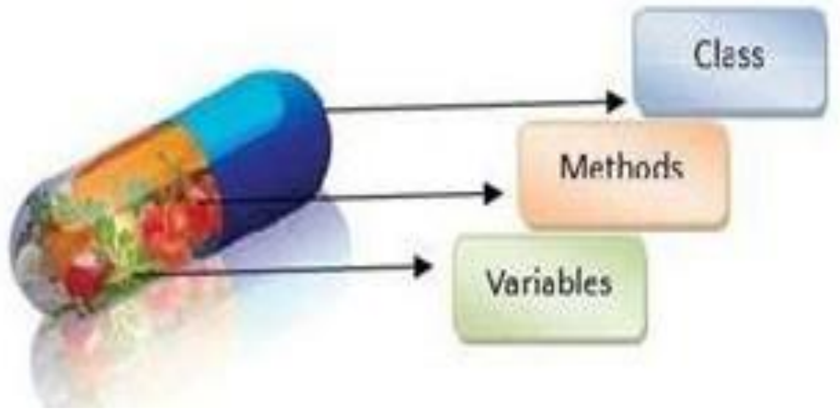
Encapsulation

- In simple words, “Encapsulation is a process of binding data members and member functions into a single unit”.



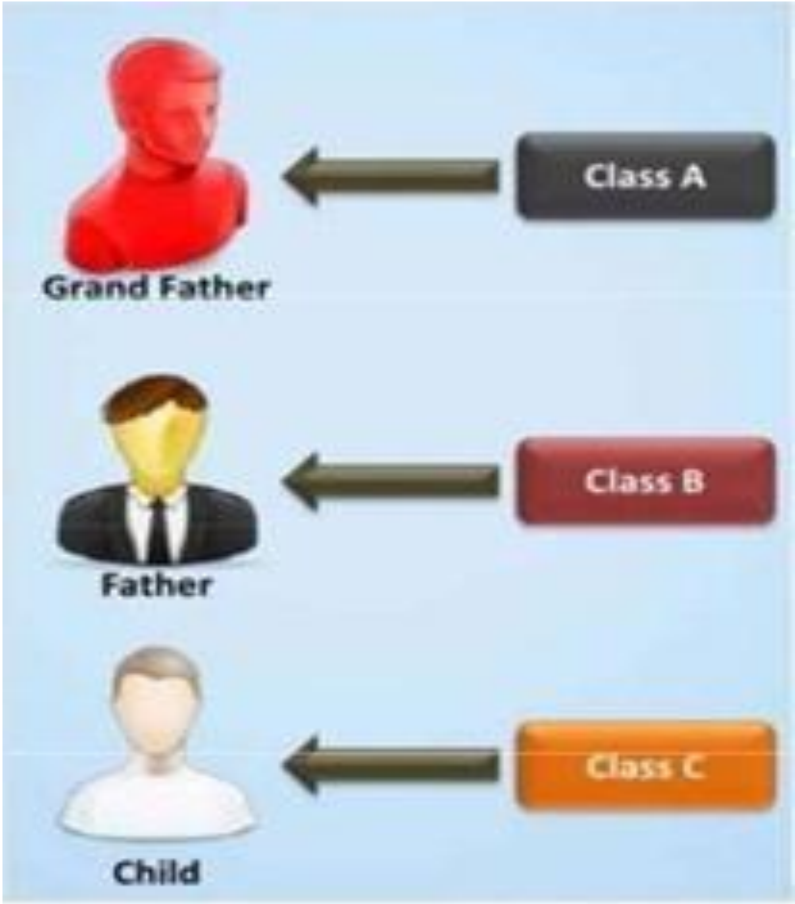
Encapsulation

- The data is not accessible to the outside world, and only those functions which are wrapped in the class can access it.
- This insulation of the data from direct access by the outside program is called data hiding or information hiding.
- A Class is the best example of encapsulation.



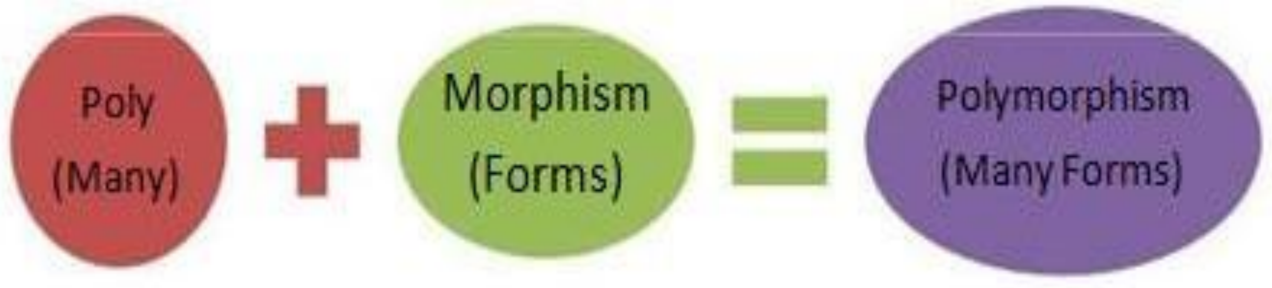
Inheritance

- Inheritance is the process by which object of one class acquire the properties of another class.
- In OOP, the concept of inheritance provides the idea of reusability.
- This means that we can add additional features to an existing class without modifying it.
- Through effective use of inheritance, you can save a lot of time in your programming and also reduce error.



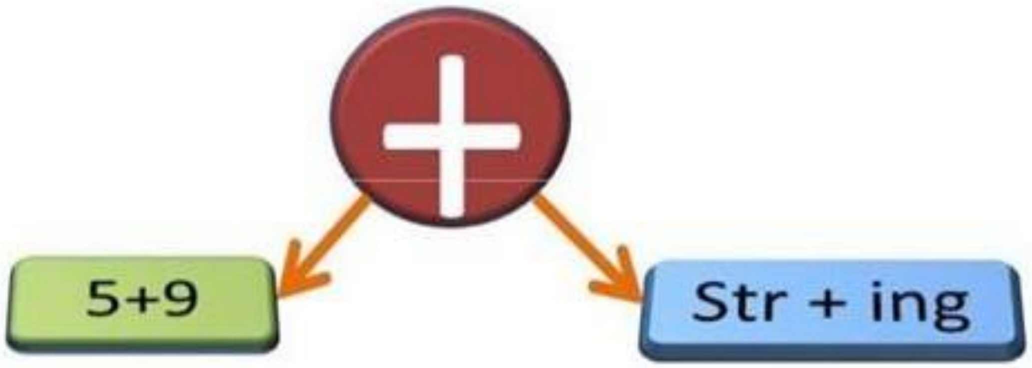
Polymorphism

- Polymorphism is a Greek term which means the ability to take more than one form.
- In polymorphism an operation may show different behavior in different instances.



Polymorphism

- For example, + is used to make sum of two numbers as well as it is used to combine two strings.
- This is known as operator overloading because same operator may behave differently on different instances.



Polymorphism

- Same way functions can be overloaded.
- For example, `sum()` function may take two arguments or three arguments etc.

    - i.e. sum (5, 7) or sum (4, 6, 8).

Dynamic Binding

- Dynamic Binding is the process of linking of the code associated with a procedure call at the run-time”.
- Dynamic binding means that the code associated with a given procedure call is not known until the time of the call at run time.

Thank You..